

PROGRAM :4

Step 1 : Create a folder prg4

```
rv24mc028_chethanacs@Elite:~/DEVops$ mkdir prg4
rv24mc028_chethanacs@Elite:~/DEVops$ cd prg4
```

Step2 :

Create index.js file

```
rv24mc028_chethanacs@Elite:~/DEVops/prg4$ nano index.js
rv24mc028_chethanacs@Elite:~/DEVops/prg4$ nano Dockerfile

const express=require("express");
const app=express;
const port=3000;

app.get("/",(req,res)=>{
app.send("hello");
});

app.listen(port,()=>{
console.log("Server running successfull at https://localhost:3000");
});
```

Step 3: Create Docker File

```
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step 4 : Install npm , package.json and package-lock.json will be created .

Step 5 : install express

```

rv24mc028_chethanacs@Elite:~/DEVops$ mkdir prog4
rv24mc028_chethanacs@Elite:~/DEVops$ cd prog4
rv24mc028_chethanacs@Elite:~/DEVops/prog4$ ls
rv24mc028_chethanacs@Elite:~/DEVops/prog4$ npm init -y
Wrote to /home/rv24mc028_chethanacs/DEVops/prog4/package.json:

{
  "name": "prog4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

rv24mc028_chethanacs@Elite:~/DEVops/prog4$ nano index.js
rv24mc028_chethanacs@Elite:~/DEVops/prog4$ npm install express

added 65 packages, and audited 66 packages in 1s

19 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
rv24mc028_chethanacs@Elite:~/DEVops/prog4$ ls
index.js  node_modules  package.json  package-lock.json
rv24mc028_chethanacs@Elite:~/DEVops/prog4$ nano Dockerfile

```

Step 6: Open Git and create a Repository

github.com/new

Install Ask Google

Apps

New repository

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * chethana-cs / **Repository name *** PROGRAM 4

✔ Your new repository will be created as PROGRAM-4.
The repository name can only contain ASCII letters, digits, and the characters ., -, and _.

Great repository names are short and memorable. How about [special-doodle?](#)

Description

LAB PROGRAM DEVOPS

18 / 350 characters

2 Configuration

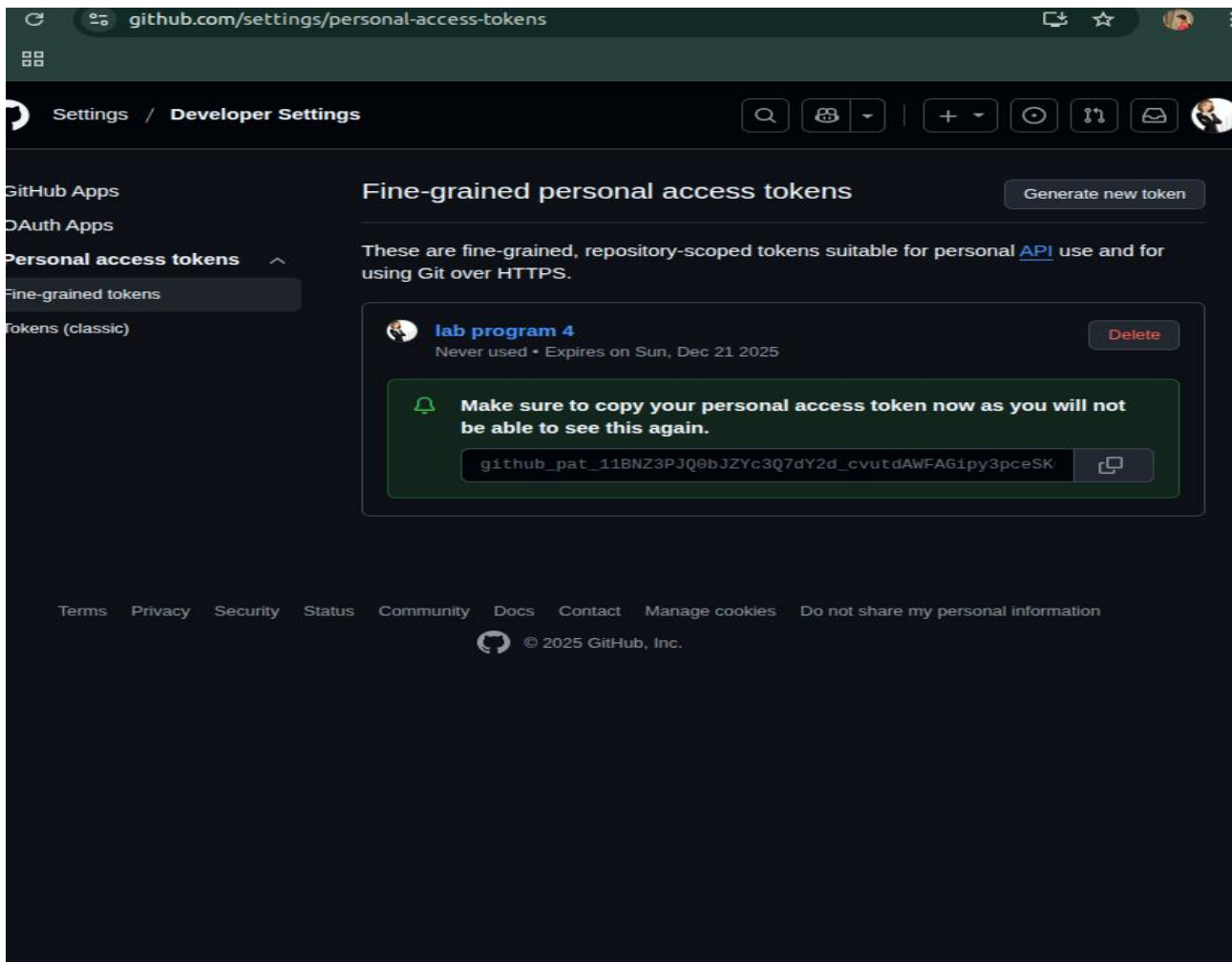
Choose visibility * Public
Choose who can see and commit to this repository

Add README On ☒
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license
Licenses explain how others can use your code. [About licenses](#)

Step 7 : In Bash enter ur Created PTA and USERNAME , Enter the below command s to add , commit and branch to to main barnch



Step 8 : to integrate with git ,

Run:

```
git init
```

Then add all files:

```
git add .
```

Check status:

```
git status
```

Commit files:

```
git commit -m "Initial commit"
```

Rename master → main:

```
git branch -M main
```

[git remote add origin https://github.com/chethana-cs/PRG4.git](https://github.com/chethana-cs/PRG4.git)

```

rv24mc028_chethanacs@Elite:~/DEVops/prog4$ git commit -m "Intial commit"
[master (root-commit) 24fdf2b] Intial commit
589 files changed, 65831 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js

```

Step 9 : to push in the git

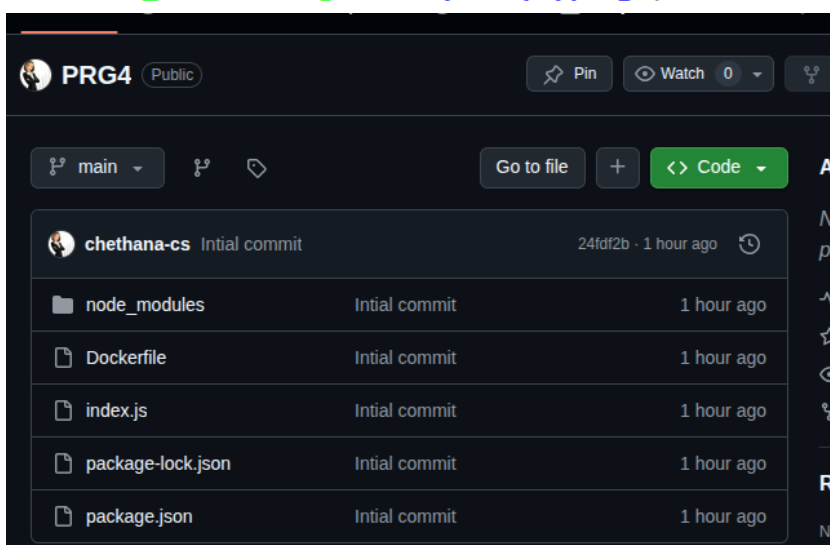
Git will ask:

- **Username** : your GitHub username
- **Password** : paste the PAT token

```

rv24mc028_chethanacs@Elite:~/DEVops/prog4$ git push -u origin main
Username for 'https://github.com': chethana-cs
Password for 'https://chethana-cs@github.com':
Enumerating objects: 677, done.
Counting objects: 100% (677/677), done.
Delta compression using up to 8 threads
Compressing objects: 100% (626/626), done.
Writing objects: 100% (677/677), 698.91 KiB | 10.28 MiB/s, done.
Total 677 (delta 117), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (117/117), done.
To https://github.com/chethana-cs/PRG4.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
rv24mc028_chethanacs@Elite:~/DEVops/prog4$

```



PROGRAM 5 :

Step 1 : Create a folder AND check version of docker and docker-compose

```
rv24mc028_chethanacs@Elite:~/DEVops/prg4$ cd ..  
rv24mc028_chethanacs@Elite:~/DEVops$ mkdir prg5  
rv24mc028_chethanacs@Elite:~/DEVops$ cd prg5  
rv24mc028_chethanacs@Elite:~/DEVops/prg5$ docker --version  
Docker version 28.5.1, build e180ab8
```

Step 2: Create a Dockerfile

Step 3 :create a Docker_compose.yml

Step 4 : run in localhost:8080

Step 5 : List the images and remove the image